Surface mass balance inference based on a transient glacier evolution model inversion.

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ABSTRACT. The distributed surface mass balance (SMB) of mountain glaciers is a key forcing for glacier evolution and subsequent meltwater availability projections. Yet, direct in situ observations cover less than 0.2 % of the world's glacierised area. Here, we present a new SMB inference method, implemented within the open-source Instructed Glacier Model (IGM), that recasts the problem as a fully differentiable transient inversion. The inversion scheme uses observed surface velocities to initialize a dynamically consistent glacier model, then inverts multi-temporal surface elevation changes (DEMs) to estimate the spatial distribution of surface mass balance that best explains the observed glacier evolution. We apply the method to three glaciers in the Alps—Argentière, Great Aletsch and Rhône—evaluating all three with a single, glacier-independent pipeline in which the basal sliding coefficient is fixed by an ensemble search on the surface-velocity misfit alone, and validate the inferred SMB against in situ stake measurements. A sensitivity test shows that this velocity misfit bounds the inferred SMB on stake-free glaciers. Our method is generic, computationally efficient and requires no glacier-specific tuning, making it readily transferable to any glacier with accurate multi-temporal DEMs and surface velocities, opening a path toward regional-scale, continuous SMB inferences.

INTRODUCTION

Mountain glaciers are sensitive recorders of climate and a critical component of regional water resources, yet projecting their evolution requires their distributed surface mass balance (SMB)—the spatial pattern of accumulation and ablation—which remains poorly constrained at the global scale, particularly in remote parts of the world. This is, in part because it relies on direct, in situ measurements using stake networks and snow pits that are labour-intensive, and hazardous in steep accumulation areas. As a result, it is currently deployed on less than 0.2 % of the world's glacierised area (Cogley and others, 2011). As an alternative, regional and global climate models provide continuous gridded SMB, but they require field calibration (Radić and others, 2014; Marzeion and others, 2012). Most glaciers therefore remain effectively unmonitored, and regional-to-global mass-balance assessments depend on extrapolation from a few benchmark sites or on model reconstructions (Hugonnet and others, 2021; Zemp and others, 2019). As remote-sensing products of glacier surface velocity (Millan and others, 2022), ice thickness (Farinotti and others, 2019; Millan and others, 2022) and elevation change (Hugonnet and others, 2021; Beraud and others, 2022) reach near-global coverage, the question is no longer whether the data exist, but how to combine them to infer physically consistent SMB fields.

Existing approaches

SMB can be derived directly from climate models that simulate surface energy balance and precipitation, but their coarse resolution and the simplified treatment of high-mountain microclimate bias precipitation phase, accumulation gradients and melt, so they rarely reproduce observed glacier change without additional bias correction (Marzeion and others, 2012; Rounce and others, 2020). Alternatively, sparse direct measurements are extended to distributed fields by homogenising them against decadal geodetic volume change, as in the long-running Glacier Monitoring Switzerland programme (GLAMOS; **?**); although these yield a valuable observational reference, they remain affected by a large number of statistical assumptions.

The route most closely related to the present work exploits the mass-conservation equation, which relates the rate of surface-elevation change to the divergence of the ice flux and the SMB (Cuffey and Paterson, 2010). Given ice thickness and surface velocity, the vertically-integrated flux can be computed, differentiated spatially, and combined with an observed elevation-change field to provide an SMB estimate. From early flux-gate estimates of emergence velocity in the ablation zone (Berthier and Vincent, 2012; Brun and others, 2018), the approach has been generalised to fully distributed SMB over entire glaciers (Bisset and others, 2020; Miles and others, 2021; Pelto and Menounos, 2021; Van Tricht and others, 2021; Kneib and others, 2022). Its state-of-the-art realisation is that of Kneib and others (2024), who reconstructed a high-resolution distributed SMB for Argentière Glacier (French Alps) over 2012–2021 from Pléiades elevation change and velocity, ground-penetrating-radar thickness and three ice-thickness models.

Two difficulties beset this route. First, the flux divergence is the spatial derivative of the product of two independently-measured noisy fields, so it is dominated by small-scale noise, which can be suppressed by *post hoc* spatial filtering (Van Tricht and others, 2021; Bisset and others, 2020); yet smoothing a quantity that respects mass conservation locally redistributes mass between grid cells and violates the very balance the method enforces. Second, and more fundamentally, the flux divergence is computed at a single observational snapshot and assumed representative of the whole observation period, whereas glacier geometry, velocity and flux all evolve under ongoing climate forcing (Jóhannesson and others, 1989; Zekollari and Huybrechts, 2015); treating the flux as stationary introduces a bias that grows with the length of the window—a limitation Kneib and others (2024) acknowledge explicitly. Per-glacier tuning of smoothing length scales, density assumptions and weighting schemes further hampers transfer to the regional scales most relevant for water-resource applications.

This study

Two features distinguish our formulation from flux-divergence inversions. First, the SMB is recovered by solving the forward mass-conservation equation rather than by differentiating observed fields: the flux divergence is recomputed at every time step from the evolving glacier state, so the inferred SMB is dynamically consistent over the whole window—rather than assuming a stationary flux—and mass is conserved exactly at every step, with no *post hoc* smoothing of a noisy divergence field. Second, we parametrise the SMB as a one-dimensional function of surface elevation, evaluated at every grid cell to yield a 2D field; this reflects the first-order elevation control on alpine SMB and reduces the inversion to a handful of unknowns, rendering the otherwise severely ill-posed problem well-constrained without site-specific smoothing length scales or density assumptions (Sect.). The price—horizontal variability within an elevation band, such as the avalanche-fed hotspots resolved by Kneib and others (2024), is not recovered—is a deliberate trade-off of sub-elevation detail for a transferable, tuning-free inversion, to which we return in Section .

The accuracy of our geometry-based inversion is ultimately controlled by that of the input DEMs. Near-global elevation-change products (Hugonnet and others, 2021) make the method broadly applicable, but their inaccuracies limit the fidelity of the recovered SMB; dedicated multi-temporal DEMs deliver the highest accuracy, and their rapidly expanding archive is what makes a regional-scale application realistic. We validate the method with two-epoch DEMs on three well-monitored Alpine glaciers of contrasting size—Argentière (French Alps), Great Aletsch and Rhône (Swiss Alps)—all inferred by the *same* two-step pipeline, without per-glacier retuning: the basal sliding coefficient is reduced to a single scalar fixed by an ensemble search on the surface-velocity misfit, and no thickness survey is assumed.

Section details the two-step pipeline—the input and validation data (Sect.), data assimilation of the initial dynamical state, then SMB inference by automatic differentiation through the transient forward run; Section presents the inferred SMB profiles and their validation against in situ stake measurements; and Section discusses the implications for regional-scale SMB inversion.

METHODS

Overview

We recover the surface mass balance (SMB) of a glacier with a two-step pipeline, implemented as a process module within the open-source Instructed Glacier Model (IGM; Jouvet, 2023; Jouvet and Cordonnier, 2023). The forward problem is governed by depth-integrated mass conservation,

$$\frac{\partial H}{\partial t} = b - \nabla \cdot \mathbf{q}, \qquad \mathbf{q} = H\,\bar{\mathbf{u}}(H, z_s, \tau_{\mathrm{ref}}), \tag{1}$$

where H is the ice thickness, b the SMB, $z_s = z_b + H$ the surface elevation, τ_{ref} the basal sliding coefficient (a reference basal shear stress; Sect.), and $\bar{\mathbf{u}}$ the depth-averaged horizontal velocity returned by IGM's deep-learning ice-flow emulator (Jouvet and Cordonnier, 2023). The emulator's network weights are pretrained on a large ensemble of higher-order flow solutions and held *frozen* throughout: it serves purely as a fast, differentiable surrogate for the ice-flow operator and is never retrained per glacier.

Equation (1) makes the central difficulty explicit: the geometry evolution depends jointly on the unknown SMB b and on the ice dynamics through the flux $\mathbf{q}$, which itself depends on H and τ_{ref}. Inferring b from an observed geometry change therefore requires the dynamical state to be fixed beforehand; otherwise an error in the dynamics would be aliased into the inferred SMB. Our pipeline reflects this dependence directly (Fig. 1): (i) we first calibrate the dynamical state by data assimilation against a surface-velocity snapshot near the start of the inversion window, inverting the ice-thickness field H; the flow is controlled not only by the geometry but also by two scalar parameters that enter the flux—the sliding coefficient τ_{ref} and the Arrhenius (rate) factor A—which we fix beforehand by an ensemble search (Sect.). (ii) Holding the calibrated state fixed, we then infer the SMB: starting from the first surface DEM, we integrate (1) forward to the epoch of the second and seek the elevation-dependent SMB profile whose simulated geometry best matches the observed one (Sect.).

Data

The pipeline requires, for each glacier, two surface DEMs bracketing the inversion window—whose difference is the geometry change to be explained—and a surface-velocity field for the data assimilation (Sect.). Ice-thickness soundings are used only where available, as an optional refinement (Appendix). We selected three well-monitored Alpine glaciers spanning a wide range of size (Table 1). For Argentière the two DEMs are the 2012 and 2021 Pléiades models of Kneib and others (2024); for Great Aletsch (2009, 2017) and Rhône (2007, 2016) they are swisstopo geodetic models. Surface velocities are from satellite feature tracking (Millan and others, 2022) for all three.

For validation we use in situ stake balances, kept strictly independent of the inversion. Rather than compare against every raw stake reading—noisy and irregularly sampled in space and time—we retain only locations with a continuous multi-year record, average each into a single balance value (± interannual standard deviation), and evaluate the inferred profile against these averages: the stake RMSE (sRMSE) reported throughout is computed against them. This yields two such repeat cells on Aletsch and five on Rhône; on Argentière the denser GLACIOCLIM network provides the published 2012–2021 mean balance at each of 25 stakes.

Ice-dynamics calibration via data assimilation

The calibration seeks the ice-thickness field that, when passed through the ice-flow emulator, best reproduces the observed surface velocities. We use IGM's existing data-assimilation module (Jouvet, 2023) to invert for the thickness H by minimising the regularised misfit

$$\mathcal{J}_{\mathrm{DA}}(H) = \alpha_u \left\| \mathbf{u}_s(H) - \mathbf{u}_s^{\mathrm{obs}} \right\|^2 + \alpha_H \left\| H - H^{\mathrm{obs}} \right\|^2_{\Omega_{\mathrm{thk}}} + \mathcal{R}(H), \quad (2)$$

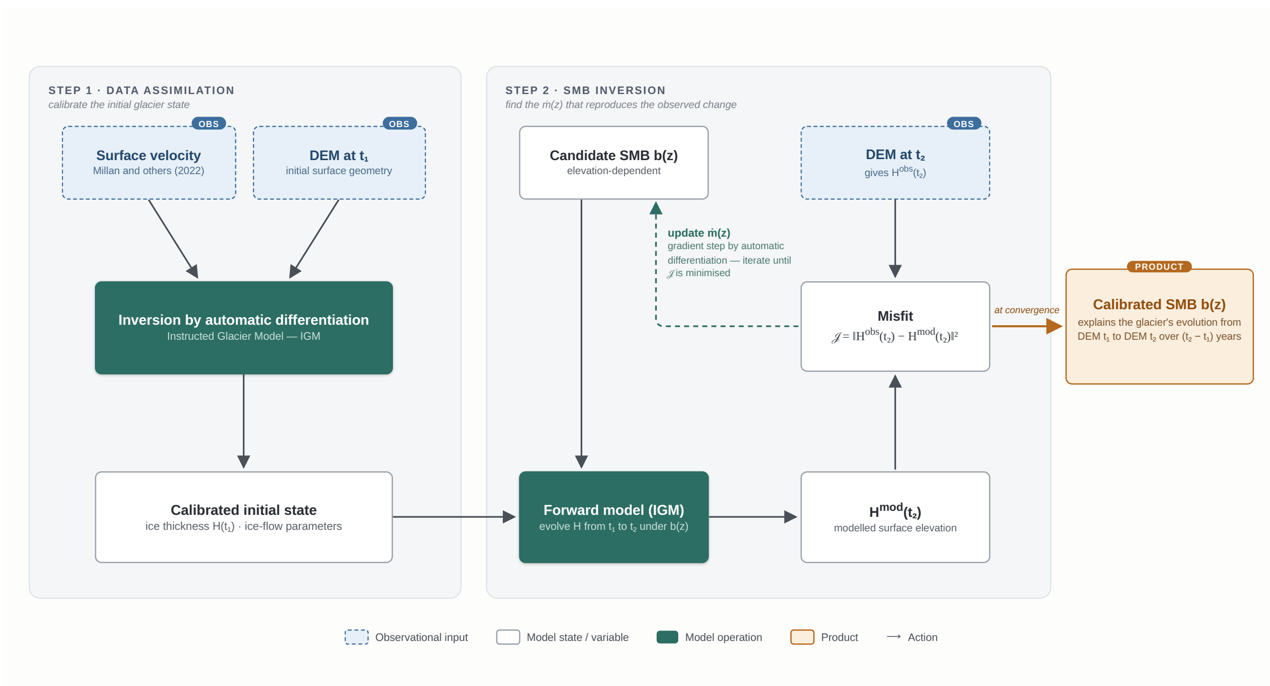

Fig. 1. Overview of the two-step SMB-inference workflow. *Step 1 (data assimilation):* the observed surface velocity and the start-epoch DEM are assimilated to recover the ice thickness H_{t_1}—and hence the bedrock—while the scalar flow parameters (τ_{ref}, A) are selected by an ensemble search that minimises the surface-velocity misfit. *Step 2 (SMB inference):* with the calibrated state held fixed, the transient model is integrated from the start DEM to the end DEM under a candidate elevation-dependent SMB profile $\beta(z)$; reverse-mode automatic differentiation through the whole forward run yields the profile whose modelled end-of-window geometry best reproduces the observed one. The ice-flow emulator (frozen weights) supplies the depth-averaged velocity at every step.

Table 1. Input and validation data for the three study glaciers. DEM epochs bracket the inversion window; velocities drive the data assimilation; GPR thickness is used only in the optional field inversion (Appendix). The stake column gives the number of continuous-record locations averaged for the sRMSE.

	Argentière	Great Aletsch	Rhône
Window	2012–2021	2009–2017	2007–2016
Surface DEMs	Pléiades	swisstopo	swisstopo
Velocity	(Millan and others, 2022)	(Millan and others, 2022)	(Millan and others, 2022)
GPR thickness	yes (app.)	–	–
Stake locations	25	2	5

where $\mathbf{u}_s(H)$ is the modelled surface velocity returned by the emulator, the second term penalises departure from point ice-thickness measurements over their survey footprint Ω_{thk} (active only where such data exist), and $\mathcal{R}$ is a Tikhonov-style smoothness regulariser on H. The minimisation is solved by L-BFGS (**?**); we refer the reader to Jouvet (2023) for the full formulation, weighting strategy and convergence behaviour. In the main analysis we assume no glacier carries a thickness survey, so the second term is dropped and the calibration is governed by the surface-velocity misfit and the regulariser alone; the smoothness weight in $\mathcal{R}$ is set at the corner of the resulting L-curve.

The emulated flow regime is not set by the geometry alone: it also depends on two scalar physical parameters, the basal sliding coefficient τ_{ref} and the ice rate (Arrhenius) factor A. Here τ_{ref} is the reference basal shear stress in Weertman's sliding law—the shear stress sustaining a reference basal speed, in MPa—so that a smaller τ_{ref} corresponds to easier sliding. Rather than invert these pointwise, we fix them beforehand by an ensemble (Optuna) search: candidate (τ_{ref}, A) pairs are each passed through the data assimilation, and we retain the pair that minimises the surface-velocity misfit, so the assimilation itself inverts the thickness H alone. We apply this identical procedure to all three glaciers, which keeps the pipeline free of glacier-specific tuning. The rate factor is held at a representative Alpine value ($A = 78\,\text{MPa}^{-3}\,\text{yr}^{-1}$) throughout, since the flow—and hence the inferred SMB—is far less sensitive to A than to τ_{ref} (Sect.).

The calibration inverts the thickness at the start epoch t_1, which we denote H_{t_1}. Fixing the thickness also fixes the bedrock,

$$z_b(\mathbf{x}) = z_s^{\text{obs}}(\mathbf{x}, t_1) - H_{t_1}(\mathbf{x}), \tag{3}$$

which, assuming z_b does not evolve over the inversion window—justified on decadal timescales—provides the fixed substrate on which the transient run subsequently evolves. Only once the calibrated state matches the observed velocity to within observational uncertainty do we accept it and proceed to the SMB inversion.

Where an independent thickness survey *is* available—as on Argentière, whose ground-penetrating-radar (GPR) soundings (Rabatel and others, 2018) make a joint inversion identifiable—the same framework can go further and relax τ_{ref} into a spatial field inverted alongside H (retaining the second term of Eq. 2). This is an optional refinement, developed in Appendix ; it is not part of the common pipeline applied to all three glaciers.

SMB inference by automatic differentiation

Observation target. With the dynamical state calibrated (Sect.), the inversion is driven by the geometry change between the two DEM epochs. The bedrock is already fixed through Eq. (3); combining it with the observed end-of-window surface DEM gives the target thickness

$$H^{\mathrm{obs}}(\mathbf{x}, t_2) = z_s^{\mathrm{obs}}(\mathbf{x}, t_2) - z_b(\mathbf{x}). \tag{4}$$

The two surface DEMs $z_s^{\mathrm{obs}}(\cdot, t_1)$ and $z_s^{\mathrm{obs}}(\cdot, t_2)$ are the glacier-specific two-epoch elevation models of Sect. ; their difference defines the geometry change the inversion must reproduce.

Forward model. With the calibrated thickness H_{t_1} and the flow scalars (τ_{ref}, A) held fixed, the glacier evolves according to (1), which we integrate explicitly with a forward-Euler scheme,

$$H^{n+1} = \max\!\big(0,\ H^n + \Delta t^n \big[\, b^n - \nabla \cdot \mathbf{q}^n \,\big]\big), \tag{5}$$

where the non-negativity projection prevents spurious negative thicknesses near the margin. The step is limited by a Courant–Friedrichs–Lewy condition, $\Delta t^n = c_{\mathrm{cfl}}\, \Delta x / \max_\Omega \|\bar{\mathbf{u}}^n\|$, with $c_{\mathrm{cfl}} = 0.2 < 1$ ensuring stability, and the flux divergence is evaluated with a centred finite-difference stencil on the staggered grid. Because $\bar{\mathbf{u}}$ is recomputed from the current (H, z_s) at every step, the flux divergence evolves with the geometry rather than being held at its initial-snapshot value—the property that removes the stationary-flux bias of geodetic flux-divergence inversions (Miles and others, 2021; Kneib and others, 2024).

SMB parametrisation. Rather than treat the SMB as an independent unknown at every grid cell—which would leave the inversion severely underdetermined, since the geometry supplies at most one constraint per cell, and none where the ice is thin or absent—we parametrise it as a one-dimensional profile

$\beta(z)$, a function of surface elevation only. This reflects the first-order elevation control on alpine SMB and collapses the unknown from $\mathcal{O}(10^4)$ pixels to $N = \mathcal{O}(10)$ elevation bins, which regularises the otherwise severely ill-posed inversion without site-specific smoothing length scales or density assumptions. The profile is discretised at N uniformly spaced elevation bins by the values $\mathbf{m} = (m_1, \ldots, m_N)$, and at each time step it is mapped to the 2D source term in (1) by evaluating it at the current surface elevation, $b(\mathbf{x}, t) = \beta(z_s(\mathbf{x}, t))$, with piecewise-linear interpolation between bins. The parametrisation is deliberately minimal; the same differentiable framework admits richer descriptions (for instance an additional aspect or debris-cover dependence) where the data warrant them, offering a principled route to lift residual non-uniqueness.

Inverse problem. We seek the profile $\mathbf{m}^\star$ that minimises the misfit between the simulated end-of-window thickness $H^{\mathrm{sim}}(\cdot, t_2; \mathbf{m})$ and the target $H^{\mathrm{obs}}(\cdot, t_2)$ from (4):

$$\mathcal{J}(\mathbf{m}) = \sqrt{\frac{1}{|\Omega|} \sum_{\mathbf{x} \in \Omega} \left(H^{\mathrm{sim}} - H^{\mathrm{obs}}\right)^2} \; + \; \mathcal{R}(\mathbf{m}), \tag{6}$$

where the first term is the RMS geometry misfit and $\mathcal{R}(\mathbf{m})$ is a Tikhonov-style smoothness penalty on the squared discrete curvature of the profile, with weight λ trading data fidelity against smoothness. The minimisation uses L-BFGS (**?**) with $M = 10$ correction pairs and a Hager–Zhang line search; the low dimensionality of $\mathbf{m}$ makes the implicit Hessian approximation accurate, and convergence is reached in $\mathcal{O}(10^2)$ function evaluations (Fig. 2). The gradient $\nabla_{\mathbf{m}} \mathcal{J}$ is obtained by reverse-mode automatic differentiation (Paszke and others, 2017) through the entire forward integration (1)–(5): every operation in the forward map—the emulator evaluation, the staggered-grid divergence, the explicit Euler update, and the piecewise-linear elevation interpolation—is a differentiable TensorFlow operation, so the chain rule applies uniformly across the forward graph and gradients propagate cleanly through every time step. The principal limitation is memory: a naïve AD trace stores the full computational graph, whose footprint scales as $\mathcal{O}(N_t \cdot N_\Omega)$ with N_t time steps and N_Ω grid points, exceeding the budget of a single GPU. We address this with gradient checkpointing (**?**): the integration is partitioned into $K = \sqrt{N_t}$ segments, only the segment-boundary states are stored during the forward pass, and the within-segment graphs are reconstructed during the backward pass, reducing the memory cost from $\mathcal{O}(N_t)$ to $\mathcal{O}(\sqrt{N_t})$ at the price of one additional forward evaluation per backward pass.

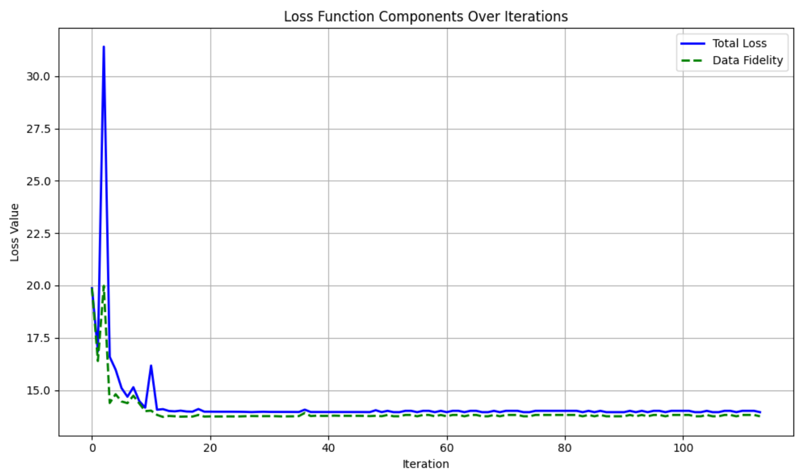

Fig. 2. Convergence of the SMB inversion on Argentière. The geometry-misfit cost (Eq. 6) against L-BFGS iteration decreases by more than an order of magnitude and levels off within $\mathcal{O}(10^2)$ function evaluations, reflecting the low dimensionality of the elevation-profile unknown $\mathbf{m}$.

RESULTS

We evaluate the method on three Alpine glaciers of contrasting size—Argentière, Great Aletsch and Rhône—through the identical two-step pipeline: for each, the data assimilation inverts the ice thickness while the sliding coefficient τ_{ref} is fixed to the scalar that minimises the surface-velocity misfit, and the SMB is then inferred by automatic differentiation. No per-glacier retuning of the regularisation, optimiser or parametrisation is performed, and none of the three uses a thickness survey. Argentière additionally carries the state-of-the-art distributed reconstruction of Kneib and others (2024), against which we compare our velocity-only result directly.

The results are presented in three parts: the calibrated dynamical states (Sect.); the inferred SMB profiles, their validation against in situ stakes, and the mass-conservation gain over stationary-flux estimates (Sect.); and the sensitivity of the inferred SMB to τ_{ref}, the dynamical parameter least constrained without thickness data (Sect.).

Calibrated ice-dynamics states

For every glacier the data assimilation converges within $\mathcal{O}(10^2)$ L-BFGS iterations to a thickness field whose emulated surface velocity matches the observations to a root-mean-square residual of a few $\mathrm{m\,yr^{-1}}$—2.4 on Argentière, 11 on Great Aletsch and 6.8 on Rhône at the adopted sliding coefficient (Table 3)—one to two orders of magnitude below typical trunk velocities (locally exceeding 100–150 $\mathrm{m\,yr^{-1}}$). The calibrated

Table 2. Validation of the inferred SMB against the GLACIOCLIM stake network on Argentière Glacier (main trunk + Améthystes tributary) over 2012–2021. The three baselines are the modelling approaches of Kneib and others (2024), evaluated against the same stakes; all three use ice-thickness data, whereas our result uses surface velocity only. The GPR-assisted field inversion, which improves further on these, is reported in Appendix .

Method	RMSE [m w.e. yr^{-1}]
F2019 (Kneib and others, 2024)	0.50
SIA (Kneib and others, 2024)	0.67
IGM-distributed (Kneib and others, 2024)	0.85
Ours (velocity only, scalar τ_{ref})	**0.65**

state fixes the bedrock through Eq. (3) and provides the starting geometry H_{t_1} for the transient forward run (Sect.). Integrated forward for one year under zero SMB it produces no spurious thickness drift, confirming that it is in mechanical equilibrium and that any subsequent geometry change is attributable to the prescribed SMB rather than to residual model imbalance.

Inferred SMB and validation

Holding the calibrated state $(H_{t_1}, \tau_{\mathrm{ref}}, A)$ fixed, we run the transient forward simulation over each glacier's window and invert for the elevation-resolved SMB profile $\beta(z)$ that minimises the geometry-misfit cost (Eq. 6). On all three glaciers the retrieved profile increases monotonically across the full altitudinal range, from strongly negative values at the tongue to a positive accumulation regime in the upper basin.

On Argentière (2012–2021) the profile reproduces the observed altitudinal pattern of the GLACIOCLIM stakes with a root-mean-square error of 0.65 m w.e. yr^{-1} (Fig. 3, Table 2). This is obtained with the generic velocity-only pipeline—no thickness survey, no per-glacier tuning, no *post hoc* smoothing—yet it is on par with the best physically-based reconstructions of Kneib and others (2024) on the same stakes, whose SIA and IGM-distributed approaches give 0.67 and 0.85 m w.e. yr^{-1} and whose flux-gate F2019 estimate gives 0.50, all of which additionally require ice-thickness data. Argentière is also covered by accurate GPR thickness soundings; repeating the experiment with them—inverting the sliding coefficient as a spatial field—lowers the error further and surpasses every published reconstruction on this site, as detailed in Appendix . Because the inversion variable is one-dimensional, horizontal heterogeneity within an elevation band—notably the avalanche-fed accumulation hotspots reported by Kneib and others (2024)—is not resolved; this deliberate trade-off, which buys a tuning-free and transferable inversion, is taken up in Sect. .

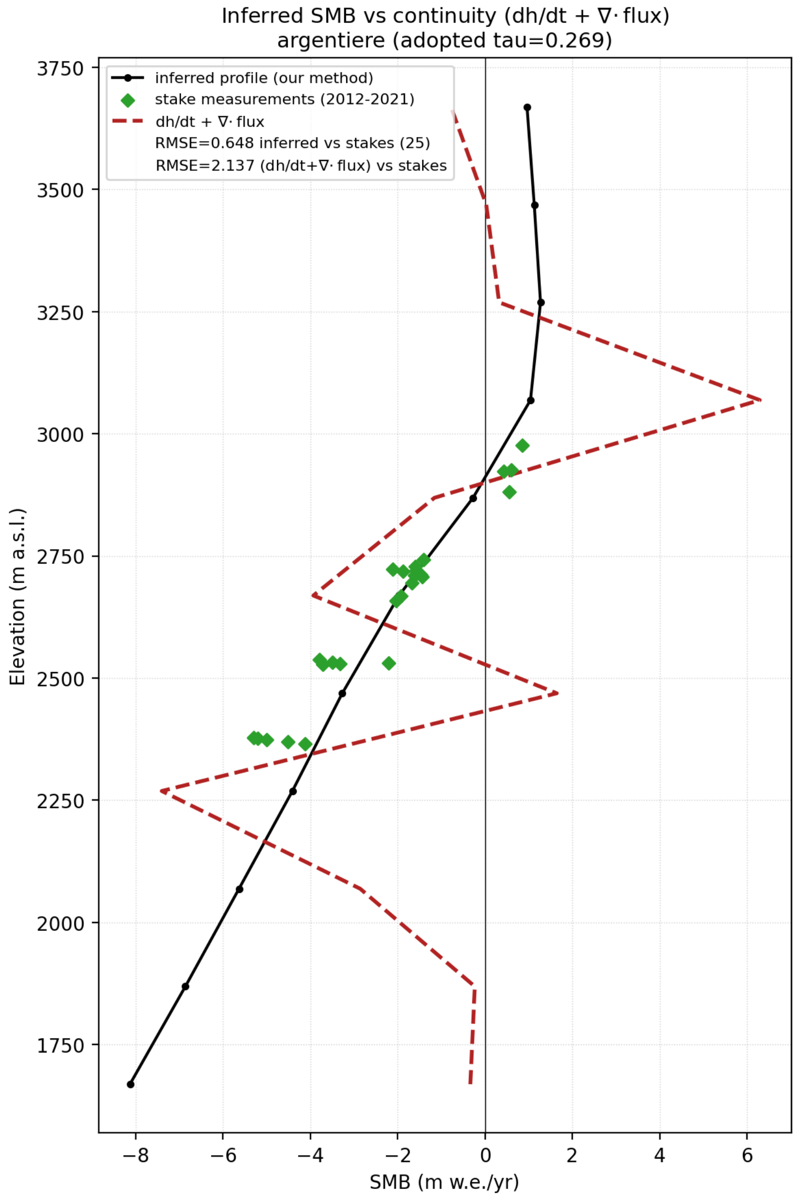

Fig. 3. Inferred elevation-resolved SMB profile for Argentière Glacier over 2012–2021 (black), obtained with the generic velocity-only pipeline ($\tau_{\text{ref}} = 0.27\,\text{MPa}$, no thickness data). Green diamonds are the 2012–2021 mean GLACIOCLIM stake balances, plotted at the elevation of each stake; the inferred profile reproduces them with $\text{RMSE} = 0.65\,\text{m w.e. yr}^{-1}$. The dashed red curve is the SMB obtained from the continuity equation with a *stationary* data-assimilation flux divergence (as in geodetic flux-divergence inversions); it departs strongly in the ablation zone ($\text{RMSE} = 2.14$), quantifying the stationary-flux bias that the transient inversion removes.

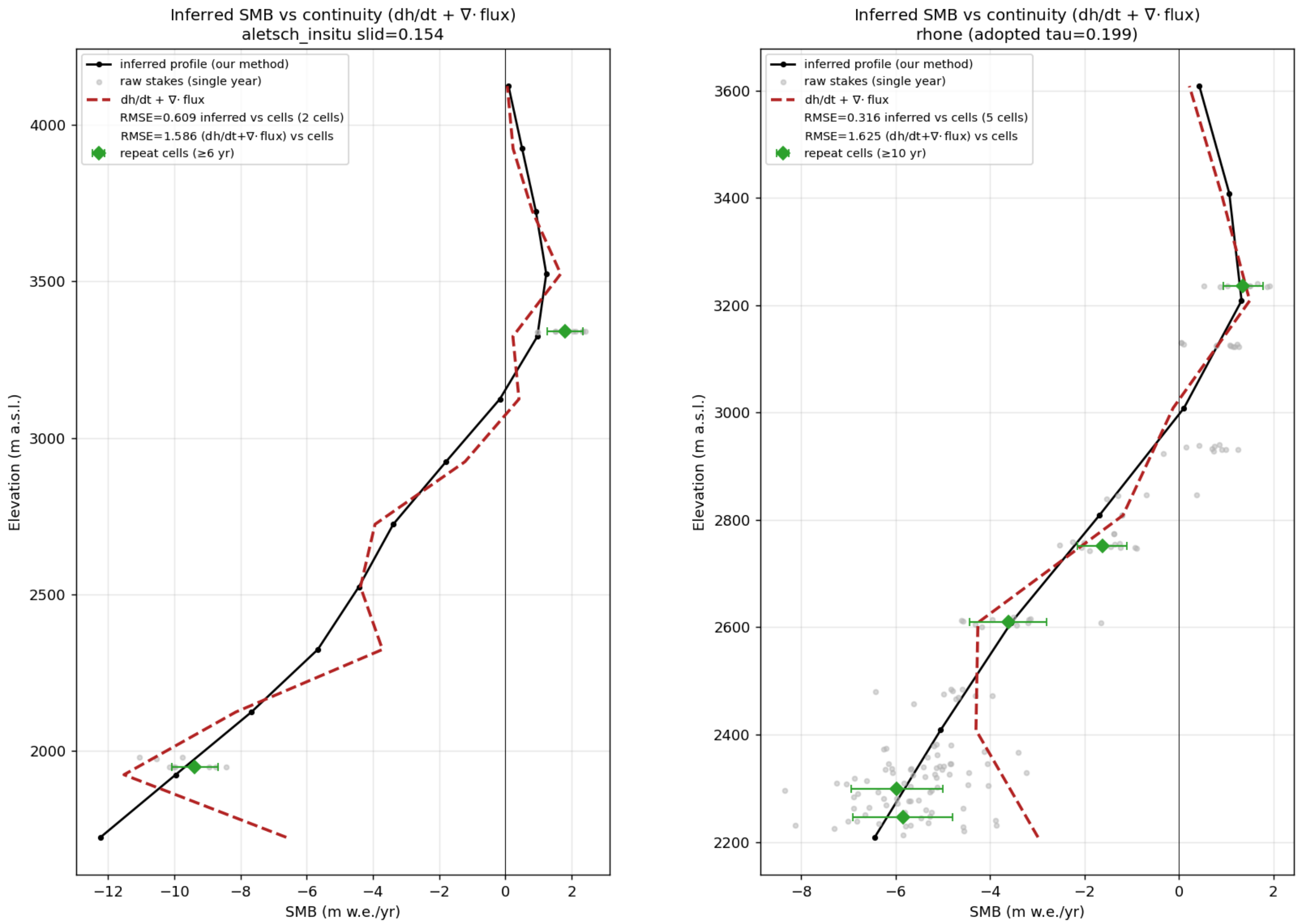

Fig. 4. Inferred SMB profile (black) for (a) Great Aletsch and (b) Rhône, compared against in situ stake measurements and against the continuity estimate $\mathrm{d}h/\mathrm{d}t + \nabla\cdot\mathbf{q}$ (dashed) obtained with the *stationary* data-assimilation flux divergence. Faint grey dots are raw single-year stake readings; green diamonds are the continuous-record averages ($\pm$ interannual standard deviation), the validation targets. The inferred profile tracks the averages across the whole elevation range (RMSE = 0.61 and 0.32 m w.e. yr^{-1}), whereas the stationary-flux estimate departs strongly in the ablation zone (1.59 and 1.63).

The same pipeline, applied without modification to Great Aletsch (2009–2017) and Rhône (2007–2016), yields equally smooth, monotonic profiles that reproduce the independent stake averages—two continuous-record locations on Aletsch and five on Rhône (Sect.)—with root-mean-square errors of 0.61 and 0.32 m w.e. yr^{-1} (Fig. 4). These are on par with the Argentière result despite the absence of any thickness constraint, which we read less as a coincidence than as evidence that the velocity-calibrated dynamical state carries the elevation structure of the SMB; whether the robustness holds across a larger and more varied sample than three large glaciers is an open question we return to in Sect. .

Figures 3 and 4 make concrete the mass-conservation argument of Sect. . The dashed curve is the SMB the continuity equation yields when the flux divergence is taken as stationary, exactly as in flux-divergence

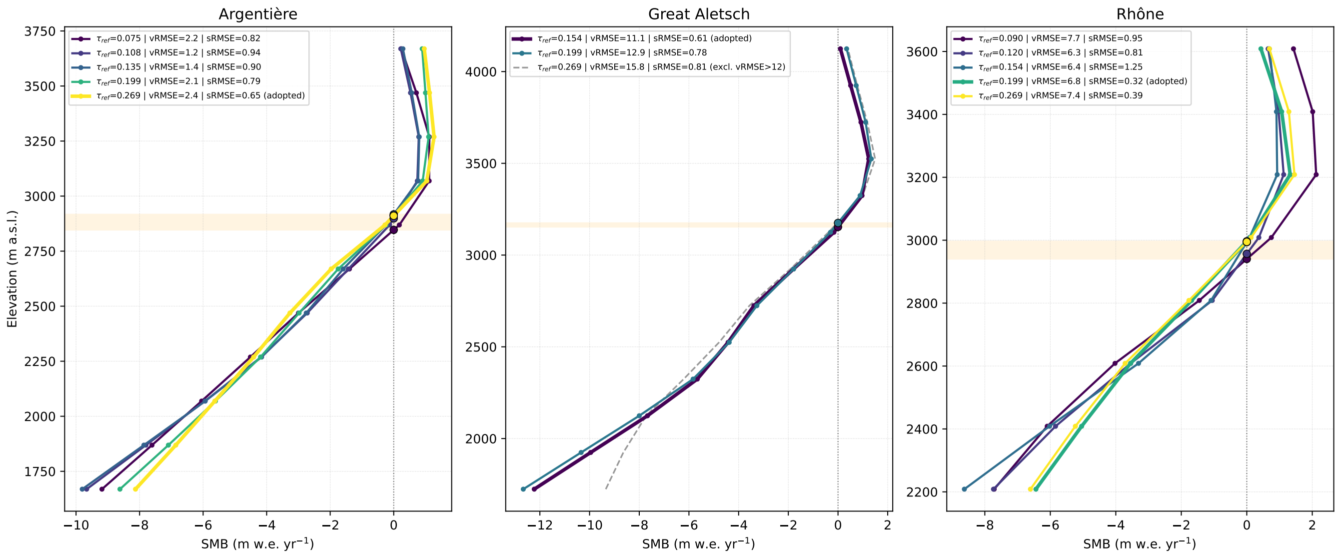

Fig. 5. Inferred SMB profiles across a range of basal sliding coefficients τ_{ref} for (a) Argentière, (b) Great Aletsch and (c) Rhône; colour encodes τ_{ref} and the adopted profile is drawn bold. Within the low-velocity-misfit basin the profiles coincide in the accumulation area and diverge only at the ablation tongue, so the residual sliding uncertainty translates into an equilibrium-line-altitude band (shaded) rather than a whole-profile bias; the high-sliding runs (dashed), which the velocity misfit rejects, depart most strongly.

inversions. On every glacier this estimate is markedly noisier than the inferred profile and diverges in the ablation zone, where the stationary divergence is least representative, raising the stake RMSE to 2.14 (Argentière), 1.59 (Aletsch) and 1.63 m w.e. yr^{-1} (Rhône)—a factor of 2.6 to 5.1 worse than our transient inversion. Because our forward model integrates the mass-conservation law directly and recomputes the flux divergence from the evolving geometry at every step, the inferred profile requires no spatial smoothing and remains mass-consistent throughout.

Sensitivity to the basal sliding coefficient

The sliding coefficient τ_{ref} is the dynamical parameter least constrained on a glacier without thickness or stake data, where its value rests on the surface-velocity misfit (*velsurf*) alone. To quantify how this choice propagates into the inferred SMB, and whether velocity data suffice to make it, we repeated the inversion across a wide range of τ_{ref} on all three glaciers (Fig. 5), using the independent stake RMSE as ground truth against the only selector available operationally.

Three results emerge. First, *velsurf* reliably rejects the corrupting regime: its worst values coincide with the high-sliding extremes, which are also where the inferred SMB is most strongly biased (Fig. 5)—so minimising the velocity misfit keeps the inversion away from the profiles that most disagree with the

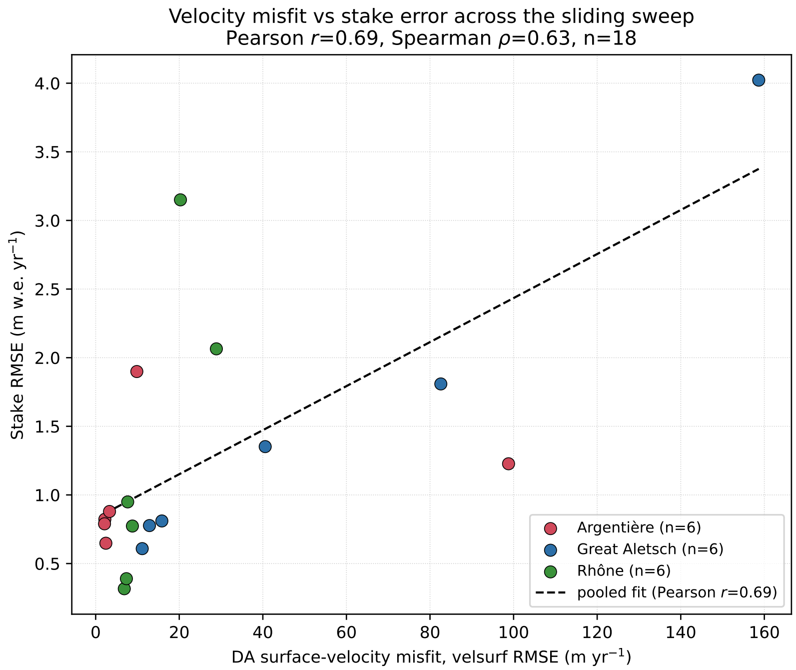

Fig. 6. Data-assimilation surface-velocity misfit (*velsurf* RMSE) against the independent stake RMSE, pooled over the sliding sweep (six runs per glacier, all three glaciers). Higher velocity misfit is associated with higher stake error (Pearson $r = 0.69$, Spearman $\rho = 0.63$), so the velocity misfit—the only selector available without stakes—carries usable information about the accuracy of the inferred SMB.

stakes. Pooling all runs across the three glaciers, higher velocity misfit is associated with higher stake error (Fig. 6; Pearson $r = 0.69$, $p < 10^{-2}$), confirming that velocity carries usable information about SMB accuracy. Second, within the low-misfit basin *velsurf* is flat and cannot pin a single optimum: the velocity-selected coefficient and the stake-optimal one are adjacent but not identical (Table 3), so the data constrain τ_{ref} to a band rather than a point; on Rhône the stake RMSE even varies non-monotonically across the basin (0.32–1.25 m w.e. yr^{-1}), confirming that velocity cannot resolve within-band differences. Third, this residual uncertainty maps almost entirely onto the ablation zone: across the flat velocity basin the accumulation-area profile is essentially unchanged, while the tongue SMB and hence the equilibrium-line altitude shift (Table 3, Fig. 5).

Taken together, these results define a velocity-only procedure for stake-free glaciers: adopt the sliding coefficient at the velocity-misfit minimum, and propagate the neighbouring coefficients within the flat basin into an SMB and equilibrium-line uncertainty band—tight in the accumulation area and widening at the tongue. The operational implications are taken up in Sect. . The rate factor A was held at the common Alpine value throughout; because it trades off with τ_{ref} in the velocity field but affects the profile far less, the band is dominated by the sliding coefficient, the weaker-constrained of the two.

Table 3. Sensitivity of the inferred SMB to the basal sliding coefficient τ_{ref} on Great Aletsch and Rhône (both at $A = 78$). *velsurf* is the data-assimilation surface-velocity misfit—the only selector available without stakes—while the stake RMSE is the independent validation against the continuous-record averages (n locations per glacier). Column minima per glacier in bold.

τ_{ref} (MPa)	velsurf RMSE ($\mathrm{m\,yr^{-1}}$)	ELA (m a.s.l.)	Stake RMSE ($\mathrm{m\,w.e.\,yr^{-1}}$)
Great Aletsch ($A = 78\,\mathrm{MPa^{-3}\,yr^{-1}}$; $n = 2$)			
0.154	**11.14**	3153	**0.61**
0.199	12.87	3175	0.78
0.269	15.84	3180	0.81
0.459	40.57	3167	1.35
Rhône ($A = 78\,\mathrm{MPa^{-3}\,yr^{-1}}$; $n = 5$)			
0.090	7.66	2940	0.95
0.120	**6.34**	2957	0.81
0.154	6.40	2994	1.25
0.199	6.84	2997	**0.32**
0.269	7.35	2995	0.39

DISCUSSION

The three experiments support a single message: a transient, differentiable inversion recovers the elevation structure of alpine SMB from geometry change and surface velocity alone, without the per-glacier tuning that flux-divergence methods require. On Argentière the generic velocity-only pipeline matches the best physically-based reconstructions of Kneib and others (2024) on the same stakes, and the optional GPR-assisted field inversion surpasses them (Appendix); on the two velocity-only Swiss glaciers the same pipeline, applied without modification, achieves comparable errors. Because the forward model integrates mass conservation directly and recomputes the flux divergence from the evolving geometry, the inferred profile conserves mass exactly and avoids the ablation-zone bias of stationary-flux estimates, which are a factor of two to five worse against the same stakes (Sect.).

The principal limitation is the one-dimensional parametrisation: horizontal variability within an elevation band—such as the avalanche-fed accumulation hotspots resolved by Kneib and others (2024)—is deliberately not recovered. This is a trade-off of sub-elevation detail for a transferable, tuning-free inversion, and it is not intrinsic to the method: the same differentiable framework admits richer descriptions of the SMB (for example an aspect or debris-cover dependence) wherever the data warrant them, providing a principled route to lift residual non-uniqueness.

Operationally, the sensitivity analysis (Sect.) defines a velocity-only recipe for stake-free glaciers: adopt the sliding coefficient at the surface-velocity-misfit minimum, and propagate the neighbouring coefficients within the flat misfit basin into an SMB and equilibrium-line uncertainty band. The robustness of the velocity-calibrated state across our three large glaciers is encouraging, but whether it persists across the far larger and more varied sample relevant to regional water-resource applications remains to be tested. Given the rapidly expanding archive of accurate multi-temporal DEMs and near-global velocity products, the tuning-free character of the method makes such a regional-scale, continuous SMB inversion a realistic next step.

APPENDIX

GPR-assisted sliding-field inversion on Argentière

The main analysis fixes the basal sliding coefficient τ_{ref} to a single scalar, so that all three glaciers are treated identically and no thickness survey is required. Where dense auxiliary data *do* exist, however, the same differentiable framework can exploit them to sharpen the dynamical state. Argentière is covered by repeated ground-penetrating-radar (GPR) soundings of ice thickness (Rabatel and others, 2018); here we show that assimilating them, and relaxing τ_{ref} from a scalar into a spatial field $\tau_{\mathrm{ref}}(\mathbf{x})$ inverted jointly with the thickness, further improves the fit and surpasses the state of the art.

The GPR soundings supply an independent thickness constraint (the second term of Eq. 2, over the survey footprint Ω_{thk}) that makes the joint inversion identifiable. The data assimilation then retrieves both the ice-thickness field H_{t_1} and the basal-sliding field $\tau_{\mathrm{ref}}(\mathbf{x})$ that together reproduce the observed surface velocity and the GPR transects, converging within 10^2 L-BFGS iterations (Fig. 7). Modelled velocities match the observations to a root-mean-square residual of $3\,\mathrm{m\,yr^{-1}}$, and the calibrated thickness departs from the GPR transects by RMS $= 12\,\mathrm{m}$ (bias $-9\,\mathrm{m}$), of the order of the GPR uncertainty itself. The recovered sliding field shows the physically expected structure—enhanced along the central flowline and reduced near the lateral margins—consistent with the finding of Kneib and others (2024) that spatial variability in the sliding coefficient helps explain the observed surface velocities.

Holding this calibrated state fixed, the transient inversion (2012–2021) yields the elevation-resolved SMB profile of Fig. 8. The simulated end-of-window thickness matches the target across the trunk to

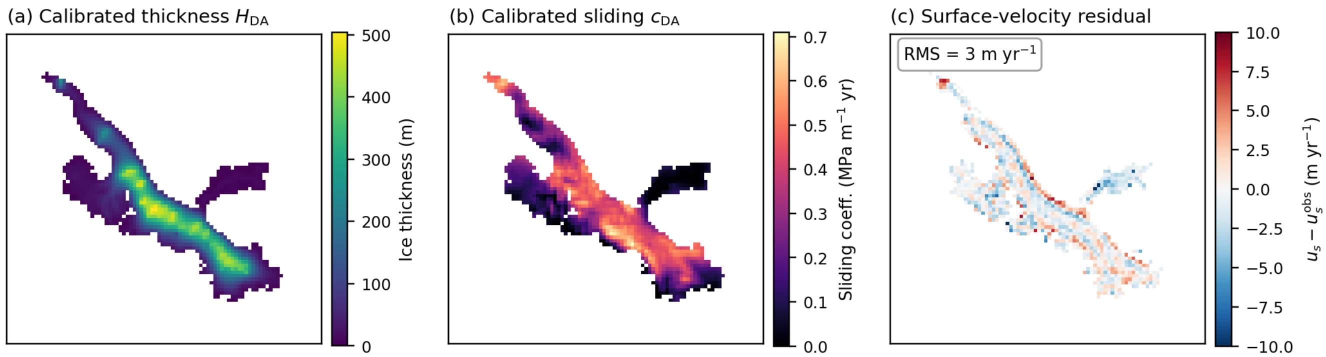

Fig. 7. GPR-assisted data-assimilation state on Argentière Glacier at the start epoch. (a) Calibrated ice thickness H_{t_1} and (b) calibrated basal-sliding field $\tau_{\mathrm{ref}}(\mathbf{x})$; (c) surface-velocity residual $u_s - u_s^{\mathrm{obs}}$, near zero across the glacier (overall RMS $= 3\,\mathrm{m\,yr^{-1}}$; white cells lack a velocity observation).

within the DEM-differencing uncertainty (Fig. 9; residual RMS $=$ 24 m, small and largely unbiased). Validated against the GLACIOCLIM network, the profile reproduces the observed altitudinal pattern with a root-mean-square error of $0.45\,\mathrm{m\,w.e.\,yr^{-1}}$—lower than the velocity-only result of the main text (0.65) and below every previously published distributed reconstruction on this site over this period, including all three baselines of Kneib and others (2024) (0.50–$0.85\,\mathrm{m\,w.e.\,yr^{-1}}$; Table 2). The residual structured misfit in Fig. 9(c) reflects horizontal variability within elevation bands—notably the avalanche-fed accumulation hotspots (Kneib and others, 2024)—that the one-dimensional profile does not represent.

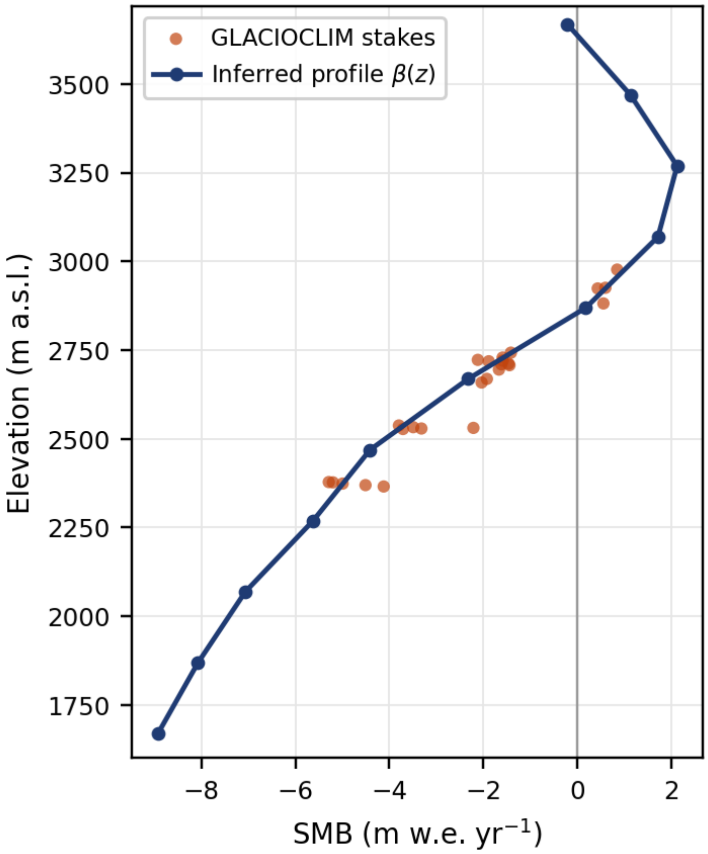

Fig. 8. Inferred elevation-resolved SMB profile $\beta(z)$ (blue) for Argentière over 2012–2021 from the GPR-assisted inversion, in water equivalent. Orange dots are the 2012–2021 mean GLACIOCLIM stake balances, plotted at the elevation of each stake. The profile spans 1670–3610 m a.s.l. and matches the stakes with RMSE $= 0.45\,\mathrm{m\,w.e.\,yr^{-1}}$.

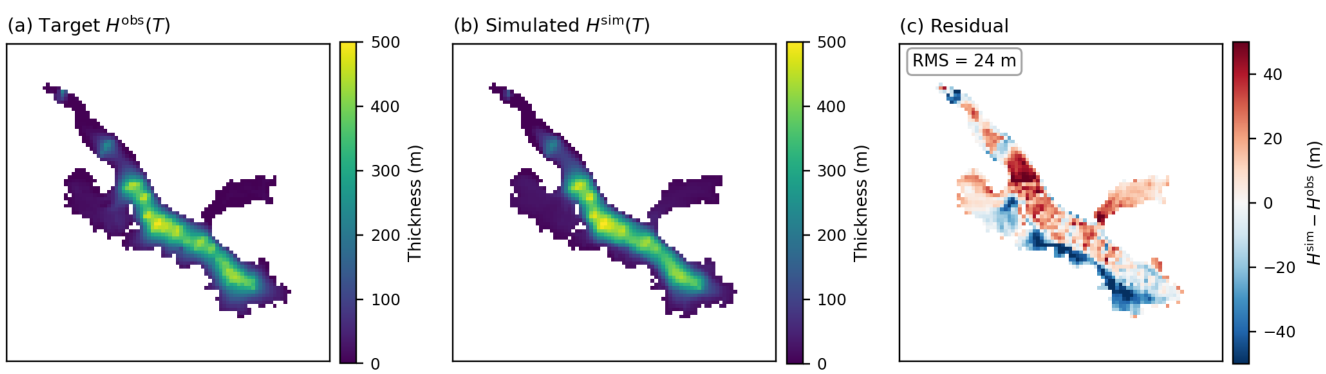

Fig. 9. End-of-window ice-thickness fit on Argentière ($t_2 = 2021$), masked to the glacier footprint. (a) Target thickness $H^{\mathrm{obs}}(\cdot, t_2)$ from the observed surface DEM and the calibrated bedrock; (b) simulated thickness $H^{\mathrm{sim}}(\cdot, t_2)$ from the forward run under the inferred SMB profile; (c) their residual $H^{\mathrm{sim}} - H^{\mathrm{obs}}$ (RMS 24 m). The residual is small and largely unbiased; its spatially structured component reflects horizontal variability within elevation bands that the one-dimensional profile does not represent.

REFERENCES

Beraud L, Cusicanqui D, Rabatel A, Brun F, Vincent C and Six D (2022) Glacier-wide seasonal and annual geodetic mass balances from Pléiades stereo images: application to the Glacier d'Argentière, French Alps. *Journal of Glaciology*, 1–13 (doi: 10.1017/jog.2022.79)

Berthier E and Vincent C (2012) Relative contribution of surface mass-balance and ice-flux changes to the accelerated thinning of Mer de Glace, French Alps, over 1979–2008. *Journal of Glaciology*, **58**(209), 501–512 (doi: 10.3189/2012JoG11J083)

Bisset RR, Dehecq A, Goldberg DN, Huss M, Bingham RG and Gourmelen N (2020) Reversed surface-mass-balance gradients on Himalayan debris-covered glaciers inferred from remote sensing. *Remote Sensing*, **12**(10), 1563 (doi: 10.3390/rs12101563)

Brun F, Wagnon P, Berthier E, Shea JM, Immerzeel WW, Kraaijenbrink PDA, Vincent C, Reverchon C, Shrestha D and Arnaud Y (2018) Ice cliff contribution to the tongue-wide ablation of Changri Nup Glacier, Nepal, central Himalaya. *The Cryosphere*, **12**(11), 3439–3457 (doi: 10.5194/tc-12-3439-2018)

Cogley JG, Hock R, Rasmussen LA, Arendt AA, Bauder A, Braithwaite RJ, Jansson P, Kaser G, Möller M, Nicholson L and Zemp M (2011) Glossary of glacier mass balance and related terms. Technical report, UNESCO-IHP, Paris

Cuffey KM and Paterson WSB (2010) *The Physics of Glaciers*. Elsevier, Amsterdam, 4 edition

Farinotti D, Huss M, Fürst JJ, Landmann J, Machguth H, Maussion F and Pandit A (2019) A consensus estimate for the ice thickness distribution of all glaciers on Earth. *Nature Geoscience*, **12**(3), 168–173 (doi: 10.1038/s41561-019-0300-3)

Hugonnet R, McNabb R, Berthier E, Menounos B, Nuth C, Girod L, Farinotti D, Huss M, Dussaillant I, Brun F and Kääb A (2021) Accelerated global glacier mass loss in the early twenty-first century. *Nature*, **592**(7856), 726–731 (doi: 10.1038/s41586-021-03436-z)

Jóhannesson T, Raymond C and Waddington E (1989) Time-scale for adjustment of glaciers to changes in mass balance. *Journal of Glaciology*, **35**(121), 355–369 (doi: 10.3189/S002214300000928X)

Jouvet G (2023) Inversion of a Stokes glacier flow model emulated by deep learning. *Journal of Glaciology*, **69**(273), 13–26 (doi: 10.1017/jog.2022.41)

Jouvet G and Cordonnier G (2023) Ice-flow model emulator based on physics-informed deep learning. *Journal of Glaciology*, **69**(278), 1941–1955 (doi: 10.1017/jog.2023.73)

Kneib M, Miles ES, Buri P, Fugger S, McCarthy M, Shaw TE, Chuanxi Z, Truffer M, Westoby MJ, Yang W and Pellicciotti F (2022) Sub-seasonal variability of supraglacial ice cliff melt rates and associated processes from time-lapse photogrammetry. *The Cryosphere*, **16**(11), 4701–4725 (doi: 10.5194/tc-16-4701-2022)

Kneib M, Dehecq A, Gilbert A, Basset A, Miles ES, Jouvet G, Jourdain B, Ducasse E, Beraud L, Rabatel A and others (2024) Distributed surface mass balance of an avalanche-fed glacier. *EGUsphere*, **2024**, 1–35 (doi: 10.5194/egusphere-2024-1733)

Marzeion B, Jarosch AH and Hofer M (2012) Past and future sea-level change from the surface mass balance of glaciers. *The Cryosphere*, **6**(6), 1295–1322 (doi: 10.5194/tc-6-1295-2012)

Miles E, McCarthy M, Dehecq A, Kneib M, Fugger S and Pellicciotti F (2021) Health and sustainability of glaciers in High Mountain Asia. *Nature Communications*, **12**(1), 2868 (doi: 10.1038/s41467-021-23073-4)

Millan R, Mouginot J, Rabatel A and Morlighem M (2022) Ice velocity and thickness of the world's glaciers. *Nature Geoscience*, **15**(2), 124–129 (doi: 10.1038/s41561-021-00885-z)

Paszke A, Gross S, Chintala S, Chanan G, Yang E, DeVito Z, Lin Z, Desmaison A, Antiga L and Lerer A (2017) Automatic differentiation in PyTorch. In *NIPS 2017 Autodiff Workshop*

Pelto BM and Menounos B (2021) Surface mass-balance gradients from elevation and ice flux data in the Columbia Basin, Canada. *Frontiers in Earth Science*, **9**, 675681 (doi: 10.3389/feart.2021.675681)

Rabatel A, Sanchez O, Vincent C and Six D (2018) Estimation of glacier thickness from surface mass balance and ice flow velocities: A case study on Argentière Glacier, France. *Frontiers in Earth Science*, **6**, 112 (doi: 10.3389/feart.2018.00112)

Radić V, Bliss A, Beedlow AC, Hock R, Miles E and Cogley JG (2014) Regional and global projections of twenty-first century glacier mass changes in response to climate scenarios from global climate models. *Climate Dynamics*, **42**(1–2), 37–58 (doi: 10.1007/s00382-013-1719-7)

Rounce DR, Hock R and Shean DE (2020) Glacier mass change in High Mountain Asia through 2100 using the open-source Python glacier evolution model (PyGEM). *Frontiers in Earth Science*, **7**, 331 (doi: 10.3389/feart.2019.00331)

Van Tricht L, Huybrechts P, Van Breedam J, Vanhulle A, Van Oost K and Zekollari H (2021) Estimating surface mass balance patterns from unoccupied aerial vehicle measurements in the ablation area of the Morteratsch–Pers glacier complex (Switzerland). *The Cryosphere*, **15**(9), 4445–4464 (doi: 10.5194/tc-15-4445-2021)

Zekollari H and Huybrechts P (2015) On the climate–geometry imbalance, response time and volume–area scaling of an alpine glacier: insights from a 3-D flow model applied to Vadret da Morteratsch, Switzerland. *Annals of Glaciology*, **56**(70), 51–62 (doi: 10.3189/2015AoG70A921)

Zemp M, Huss M, Thibert E, Eckert N, McNabb R, Huber J, Barandun M, Machguth H, Nussbaumer SU, Gärtner-Roer I, Thomson L, Paul F, Maussion F, Kutuzov S and Cogley JG (2019) Global glacier mass changes and their contributions to sea-level rise from 1961 to 2016. *Nature*, **568**(7752), 382–386 (doi: 10.1038/s41586-019-1071-0)